Mendel's scientific method

The first studies of inheritance were carried out by Gregor Mendel in 1866. He studied many characteristics of the garden pea and eventually discovered that inheritance followed precise rules. Mendel chose peas because normally pea plants are self-fertilising and their pollination can be controlled. His research involved careful planning. He used thousands of plants and experimented over eight years. Mendel became the first scientist to understand the importance of statistics in investigations and to apply mathematics to biological problems.

Mendel's first step was to spend two years growing the same peas, ensuring each flower fertilised itself. During fertilisation, pollen from stamens containing the male sperm fuse with a female egg in order to produce a seed. Every time Mendel's offspring plants looked the same as the parent plants. In this way Mendel made sure that the plants were purebred – that is, their chromosomes were homozygous.

1 How do you think Mendel could ensure that each pea flower fertilised itself?

2 Why was it important to make sure that the pea plants were purebred?

3 Use these terms to label the pea flower diagram:
eggs, ovary, petal, pistil, pollen, sepal, stamen.

Structure of a pea flower



4 Apply your own maths skills by filling in the missing numbers in the following description of Mendel's method.

Let's assume that Mendel selected 2 pea types to experiment with: a tall pea plant and a short pea plant. He didn't pay attention to any other characteristics apart from size.

While the flowers of 10 tall pea plants were still in bud, Mendel carefully opened the 2 buds of each plant and cut off the 6 stamens in each one. Altogether he opened 20 buds and removed _____ stamens. He wanted to make sure that the peas would not self-fertilise.

Then Mendel used a small brush to touch the stamens of flowers of other small pea plants, picking up pollen with the brush. He touched the pistils of the 20 flowers of the tall pea plants with the loaded brush. Then each bud was covered with a tiny, labelled bag, which could have said: 'Pollen from short peas'. In a similar manner, Mendel transferred pollen from other tall pea plant flowers to _____ flowers of short pea plants, which had had their stamens removed. Again the buds were bagged and labelled: 'Pollen from tall peas'. Altogether, Mendel wrote _____ labels for the tall and short plants.

Then Mendel waited until seeds had formed. He collected the pods from the _____ little bags and shelled them. If we assume that each pod contained 5 seeds, Mendel ended up with _____ seeds.

When the soil was prepared again, Mendel planted all those seeds. The seeds germinated and grew into plants. Mendel noticed that every one of the new pea plants was tall. Again when 2 buds on each plant had formed (_____ buds in total), he took out the stamens (altogether _____ stamens). But this time he used the pollen from those collected stamens to transfer them to the pistils (altogether 400 pistils). Like before, Mendel bagged and labelled the fertilised buds (altogether 400 labels).

At the end of the season he collected _____ pea pods and shelled them all. If we assume again that each pod contained 5 seeds, Mendel had altogether _____ seeds.

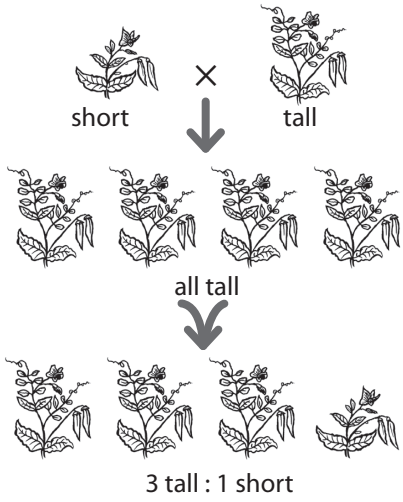
In the next season, (there are 2 pea seasons per year) Mendel planted the seeds. As the pea plants grew, he counted 1 500 tall and 500 small pea plants. He calculated the ratio of tall to small plants by dividing both numbers by the smaller one and arrived at a ratio of _____ tall plants to _____ short plant.

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From his experiments, Mendel concluded several things:

- Each purebred parent (P generation) gave only one factor to its offspring. We now call those factors *genes*.
- Since each plant in the F1 generation (children) had a factor for tallness and a factor for shortness (genotype) but looked tall (phenotype), Mendel determined that tall was dominant over short which was recessive.
- Although each tall factor and each short factor could join with either a tall or a short factor in the F2 generation (grandchildren), most of the F2 generation looked tall.



Mendel was not able to predict exactly what each F2 offspring would look like, but he was able to state that each offspring had a 75% chance for the dominant factor (tall) to appear.

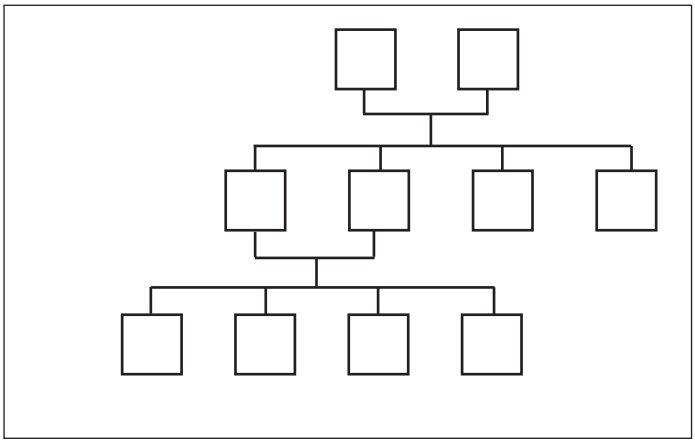
Mendel altogether looked at seven different features – size was only one of them. He learned that each feature had only two expressions; for example, the shape of the pod could be either inflated or wrinkled and the flower colour could be either purple or white. He learned that each feature was independent of every other feature, so it was possible that plants with dark-coloured pods had yellow or green seeds and white or brown seed coats.

Mendel showed that the appearance of different characteristics in heredity (phenotype) followed two laws. The **law of segregation** states that each feature is determined by two factors (characteristics, now called genes), which are either dominant or recessive, and that only one factor is passed from each parent to the offspring. The dominant factor hides the recessive factor when both factors occur together (genotype); for example, for size tall hides small. The **law of independent assortment** states that factors of traits appear independently of other traits; for example, a tall plant can have either purple or white flowers and round or wrinkled seeds.

Mendel's work was astonishing because he repeated it so many times and applied maths to his process and his results. His findings did not make sense to other scientists of his time, who believed that the parents' characteristics were mixed up in the offspring. It was more than thirty years before scientists recognised that Mendel had laid the foundation for the new discipline of genetics.

5 Use the steps below to help you label the family tree for Mendel's peas. For each plant, list the phenotype (tall or short) below the box and its genotype (pairs of genes) inside the box.

- Start with a pure bred tall plant and a pure bred small plant. This is the parent generation (P). The pollen of the small plant are used to pollinate the pistils of the flowers of the tall plant. The flowers develop four seeds which then grow into plants. This is the children generation (F1).
- Only pollen from two F1 plants are used to fertilise the pistils of their flowers. The resulting four seeds again are planted and allowed to grow into plants. This is the grandchildren generation (F2).



6 Why did Mendel use pea plants for his experiments, and not dogs or humans?

7 Mendel's experiments would have been much quicker if he had used 1 tall plant and 1 short plant, and grown three generations from them. Why didn't he?

8 Mendel was not able to predict exactly what each F2 offspring would look like. Why?

9 In what ways do you use statistics and other kinds of mathematics in science?

10 Using Mendel's method as a guide, describe the steps you think would be involved in the medical testing of a new type of drug. How would statistics and other kinds of maths be used?

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